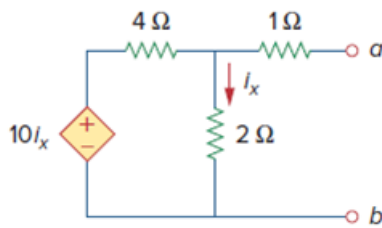


Problem

Obtain the Thevenin equivalent seen at terminals a - b of the circuit in Fig. 4.130.

Figure 4.130:



Step-by-step solution

Step 1 of 5

Refer to figure 4.130 given in the textbook.

Apply Kirchhoff's voltage law and determine the Thevenin voltage.

$$10i_x - 4i_x - 2i_x = 0$$

$$i_x = 0$$

So, the Thevenin voltage is also zero.

It can also be seen that; the given circuit does not contain any independent source.

Therefore, the Thevenin's voltage across the terminals is $V_{Th} = 0V$.

Step 2 of 5

Determine the Thevenin's resistance R_{Th} . Excite the network with a voltage source v_o connected to the terminals a and b as shown in figure 1.

Set $v_o = 1V$ to ease calculations.

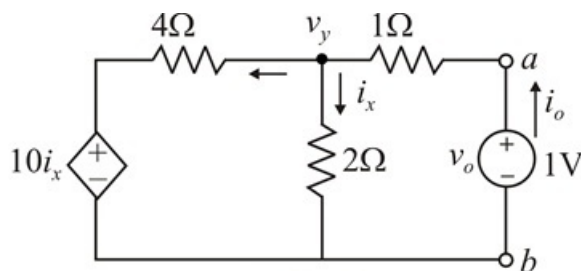


Figure 1

Step 3 of 5

Here, v_y is the voltage at node-y.

Consider the circuit shown in figure 1.

Calculate current i_x .

$$i_x = \frac{v_y}{2}$$

Apply Kirchhoff's current law at node of voltage v_y for the circuit shown in figure 1.

$$\frac{v_y - 10i_x}{4} + \frac{v_y}{2} + \frac{v_y - 1}{1} = 0$$

$$v_y - 10i_x + 2v_y + 4v_y - 4$$

$$-10i_x + 7v_y = 4 \dots\dots (1)$$

Step 4 of 5

Substitute $\frac{v_y}{2}$ for i_x in equation (1).

$$-10 \left(\frac{v_y}{2} \right) + 7v_y = 4$$

$$-5v_y + 7v_y = 4$$

$$2v_y = 4$$

$$v_y = 2 \text{ V}$$

Step 5 of 5

Calculate the current i_o through the 1 V source.

$$i_o = \frac{1 - v_y}{1}$$

Substitute 2 for v_y .

$$i_o = \frac{1 - 2}{1}$$

$$i_o = -1 \text{ A}$$

Calculate the Thevenin's resistance R_{Th} .

$$\begin{aligned} R_{Th} &= \left| \frac{1}{i_o} \right| \\ &= \left| \frac{1}{-1} \right| \\ &= 1 \Omega \end{aligned}$$

Therefore, the Thevenin's equivalent voltage V_{Th} for the given circuit is 0V and Thevenin's resistance R_{Th} is 1 Ω.